4.1. Antidiabetic Drugs

Several drugs for type 2 diabetes also act on pathways that drive inflammation in RA, such as AMP-activated protein kinase (AMPK) and NF-κB (Section 3). In RA, insulin resistance also correlates weakly with disease activity ($\rho = 0.21$) and more strongly with inflammatory cytokines such as interleukin-6 ($\rho = 0.63$) [1]. This section first covers metformin and pioglitazone, followed by glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium–glucose cotransporter-2 (SGLT2) inhibitors.

4.1.1. Metformin

Metformin is a common oral treatment for type 2 diabetes, but how it acts at the cellular level was long unclear [2]. The drug appears to act mainly by activating AMPK, which in hepatocytes lowers acetyl-CoA carboxylase activity, increases fatty acid oxidation and suppresses lipogenic enzyme expression [2]. Activated AMPK phosphorylates raptor, a step needed to inhibit mammalian target of rapamycin (mTOR) complex 1 (mTORC1) under energy stress [3]. Both mTOR complexes, mTORC1 and mTORC2, promote T helper 17 (Th17) cell differentiation [4].

Two animal studies showed that metformin reduced arthritis and protected the joint. In the collagen-induced arthritis (CIA) model (Table 1), immunized mice were given 5 mg/mouse of metformin orally three times per week for 9 weeks, starting 7 days after the first immunization. This treatment significantly reduced arthritis severity and shifted the Th17 to regulatory T (Treg) cell ratio, with fewer Th17 cells and more Treg cells. The numbers of T cells positive for phosphorylated signal transducers and activators of transcription 3 and 5 (STAT3 and STAT5) fell and rose, respectively, and osteoclast activity was reduced [5]. Similarly, in rats with CIA, intraperitoneal (i.p.) metformin (100 mg/kg every 3 days for 3 weeks) protected the knee joints on micro-computed tomography (micro-CT) and on histopathology. Micro-CT showed preserved trabecular bone, histology showed less osteoclast formation and synovitis, and metformin also lowered markers of systemic inflammation, chondrocyte apoptosis and cartilage matrix degradation [6]. After body-surface-area conversion for a 60-kg adult [7], the oral dose of 5 mg/mouse [5] equals approximately 810–1,220 mg per dose. This range assumes a 20–30-g mouse,

because body weight was not reported. The i.p. dose of 100 mg/kg in rats [6] equals approximately 970 mg per dose. These per-dose values lie within the usual clinical range of 500–2,550 mg/day [8]. Averaged over the dosing intervals of three times weekly and every 3 days, however, they correspond to approximately 350–520 mg/day and 320 mg/day, respectively, at or below the lower limit of that range [7].

Human joint tissue data also suggest a link between AMPK and RA activity. RA synovium expressed more AMPK-α1 than osteoarthritis synovium; in RA serum, phosphorylated AMPK-α1 correlated weakly with the 28-joint Disease Activity Score (DAS28; $r = 0.27$), and total AMPK-α1 correlated moderately with tumor necrosis factor-α (TNF-α; $r = 0.46$) [9]. In cultured RA fibroblast-like synoviocytes (FLS), metformin inhibited proliferation and migration at millimolar concentrations (5–10 mM) [9]. It also lowered the levels of TNF-α, interleukin (IL)-1β and IL-6 in these cells [9].

Three clinical studies suggest a benefit of metformin as an add-on to conventional synthetic (cs) disease-modifying anti-rheumatic drugs

(DMARDs). The larger of the two randomized trials, a double-blind study from Egypt, assigned 120 patients with active RA to metformin 1,000 mg or placebo daily, each added to methotrexate (MTX) 7.5 mg/week, over 12 weeks [10]. At 12 weeks, 80.8% of the metformin group reached an American College of Rheumatology 20% improvement (ACR20) response, compared with 54.7% of the placebo group (P = 0.001), without serious adverse effects in either group [10]. A randomized, single-blind trial enrolled 66 Egyptian patients with moderately or highly active RA, of whom 33 received metformin 850 mg twice daily in addition to csDMARDs and 33 received placebo. After 6 months, C-reactive protein (CRP) levels and DAS28-CRP scores were lower than with placebo among the 60 patients who completed the trial, and three metformin-treated patients (9%) withdrew because of gastrointestinal side effects [11]. Because these withdrawals were excluded from the analysis, the benefit may be overestimated. In a non-randomized prospective cohort study, 124 patients received low-dose metformin (250 mg twice daily) in addition to traditional csDMARD therapy for 90 days and were compared with 98 patients receiving

csDMARDs only. Metformin use was associated with a faster fall in IL-6, TNF-α, high mobility group box 1 (HMGB1) and CRP than in the csDMARD-only group [12]. A meta-analysis of four randomized trials with 350 patients found that metformin lowered CRP after 3 months (mean difference −5.99 mg/L, 95% confidence interval [CI] from −7.46 to −4.52) [13]. Over the same period, DAS28 also fell in the three trials that reported it (mean difference −0.80, 95% CI from −0.87 to −0.73) [13]. This interval appears implausibly narrow. The three trials were small, and DAS28 varied by about 1.5 units between patients [10]; the pooled estimate should therefore be read with caution. However, each of the four trials was conducted in Egypt or India, and two carried a high risk of bias because they were single-blind and did not report trial registration [13]. Therefore, metformin may be a well-tolerated add-on to csDMARDs for lowering RA disease activity in short trials, but larger and longer double-blind randomized controlled trials (RCTs) that also check joint damage on X-ray are needed.

Data on RA prevention are conflicting. In a large UK cohort of people recently diagnosed with type 2 diabetes, metformin use showed no clear

association with RA risk compared with other oral diabetes drugs (adjusted hazard ratio [HR] 1.00, 95% CI from 0.82 to 1.22) [14]. A Mendelian randomization study suggested that metformin may protect against RA (odds ratio [OR] 0.023, 95% CI from 0.0016 to 0.34) [15]. This effect size is implausibly large for a drug effect and should be interpreted with caution. Genetic variants linked to metformin use may also reflect the genetic liability to diabetes itself. In an Israeli cohort of 113,749 people starting metformin, adherence was associated with a lower RA risk, which was lowest when 40–59% of days were covered rather than under 20% (adjusted HR 0.62, 95% CI from 0.45 to 0.84) [8]. Because the risk was lowest at intermediate rather than the highest adherence, the association did not rise steadily with adherence, which leaves healthy-adherer bias and chance as plausible explanations. In analyses stratified by sex, this association was limited to women [8].

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